

ADVANCED IMAGE PROCESSING

ORAL AIP

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TP1 - IMAGE DENOISING

Original Image



Noisy Image (Gaussian)



Bilateral Filter (OpenCV)



Bilateral Filter (Scratch)



Objective: Compare
different image
denoising methods

Gaussian noise
(mean=0, sigma=10)

TP1 - IMAGE DENOISING

sigmaColor=5
sigmaSpace=5



sigmaColor=5
sigmaSpace=15



sigmaColor=5
sigmaSpace=30



sigmaColor=15
sigmaSpace=5



sigmaColor=15
sigmaSpace=15



sigmaColor=15
sigmaSpace=30



sigmaColor=30
sigmaSpace=5



sigmaColor=30
sigmaSpace=15



sigmaColor=30
sigmaSpace=30



sigmaSpace – Controls how much spatial distance influences the weighting.

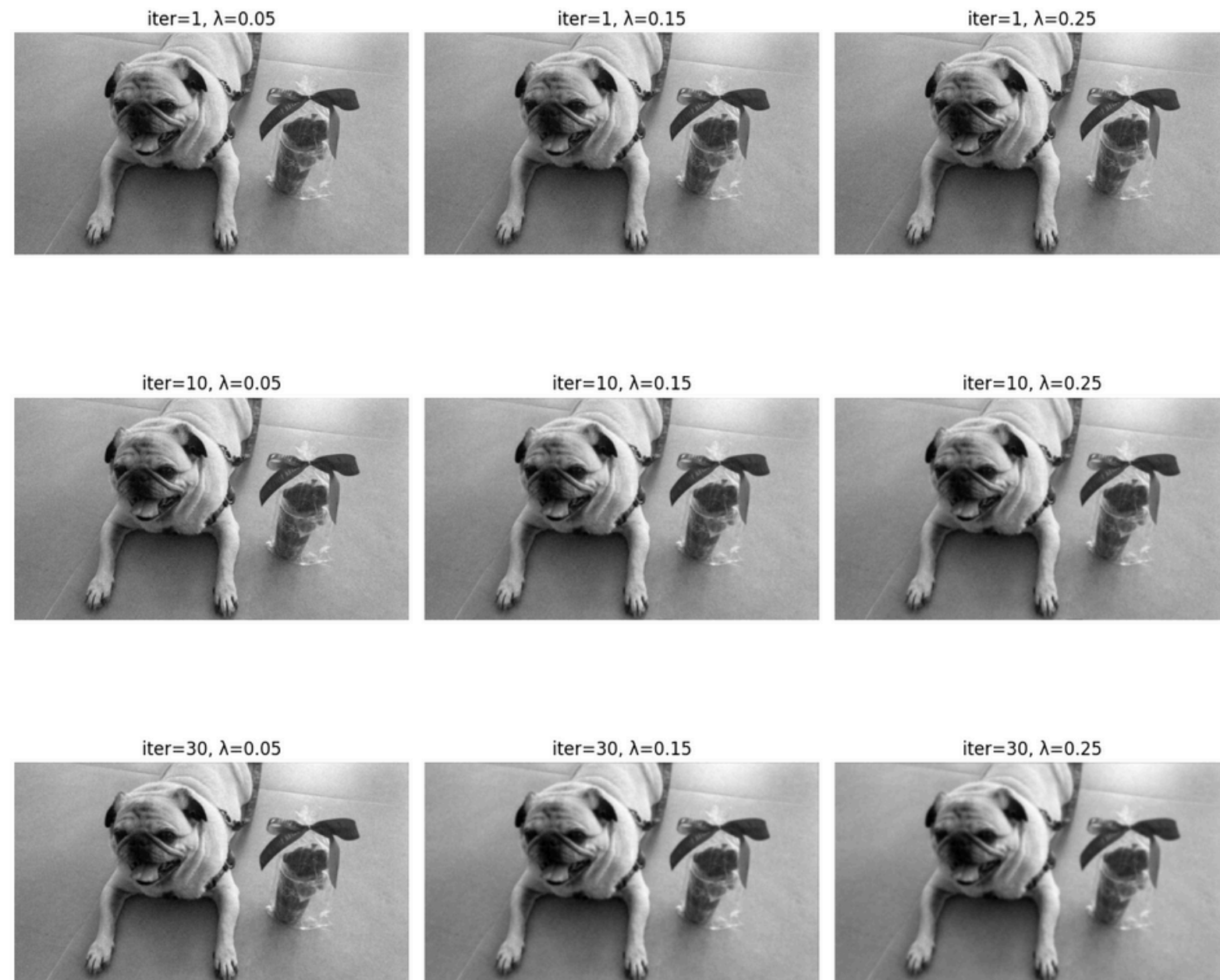
sigmaColor – Controls how much intensity difference influences the weighting.

d – Diameter of the pixel neighborhood

cv2 vs Scratch
0.4s vs 49s

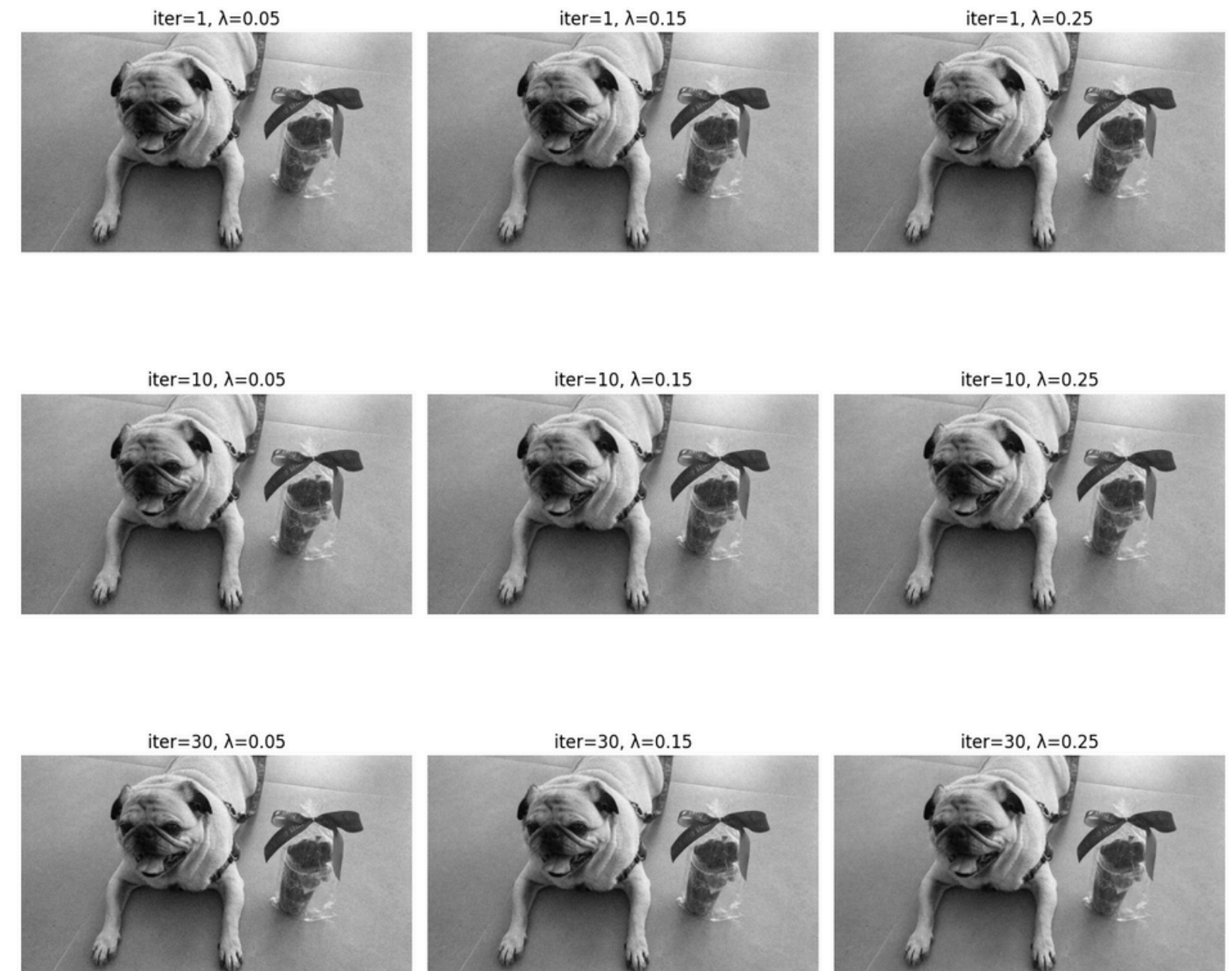
Isotropic Diffusion

Diffusion isotropique : comparaison $\text{num_iter} \times \lambda$



Anisotropic Diffusion

Diffusion anisotrope : comparaison $\text{num_iter} \times \lambda$



d_values = [3, 5, 9, 15]

sigma_values = [3, 5, 10, 15, 20, 25]

n_iter_values = [0, 1, 3, 5, 10]

lambs = [0.05, 0.15, 0.25]

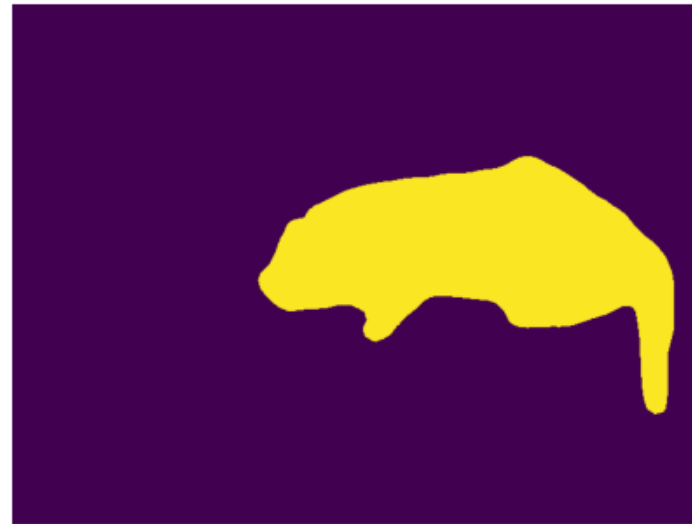
Filter	Parameters	MSE	PSNR	SSIM
Bilateral (ours)	d=5, sigma=20	26.38	33.92	0.8630
Bilateral (cv2)	d=9, sigma=20	26.88	33.84	0.8685
Gaussian	d=3, sigma=3	52.63	30.92	0.8108
Isotropic Diffusion	n_iter=5 kappa=0.05	0.0006	31.90	0.8092
Anisotropic Diffusion	n_iter=5 kappa=0.05	0.0007	31.79	0.8084

TP TRANSFORMER SEGMENTATION

Original



FCN



CLIPSeg



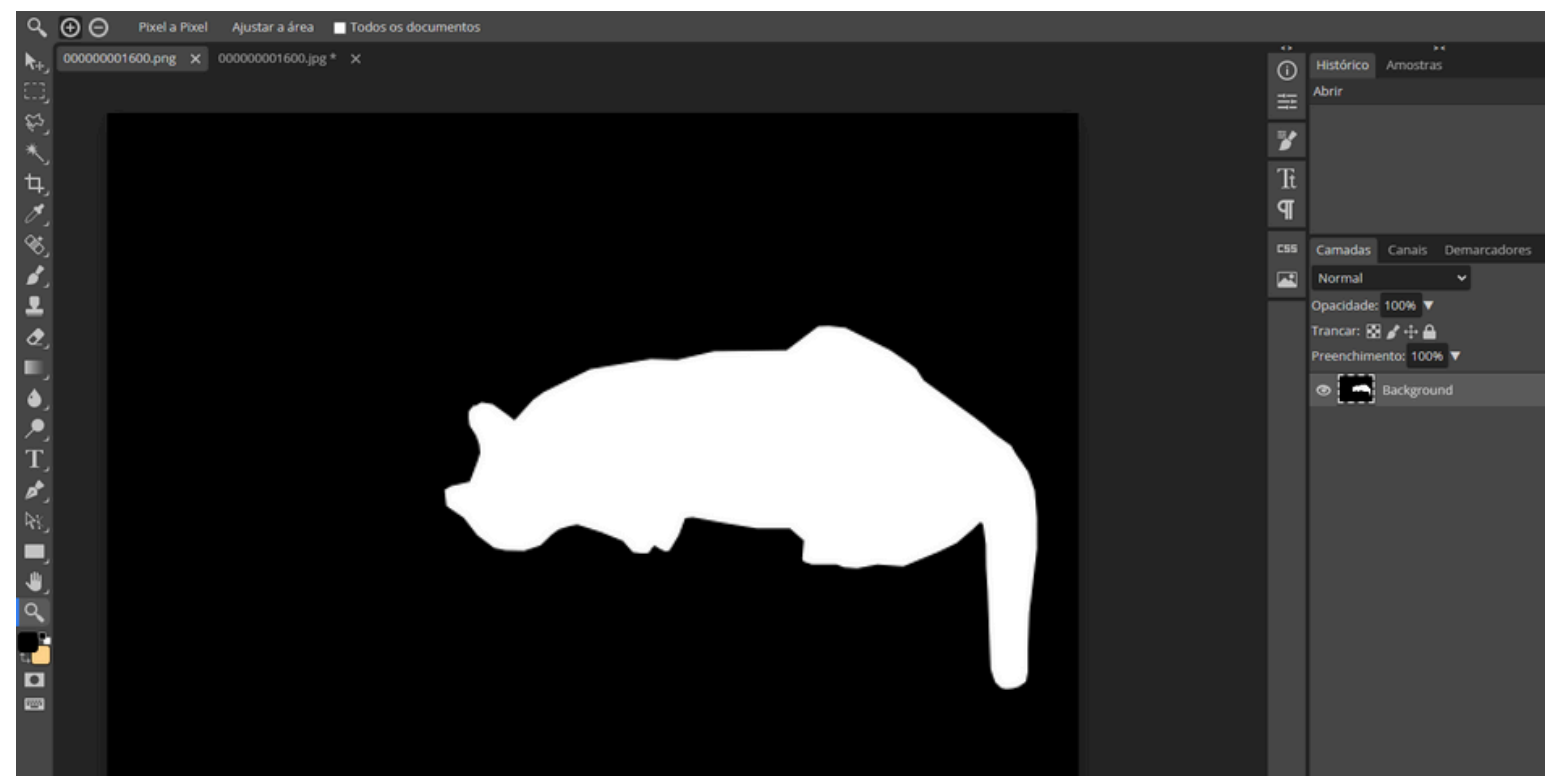
Dice

0.944



0.398

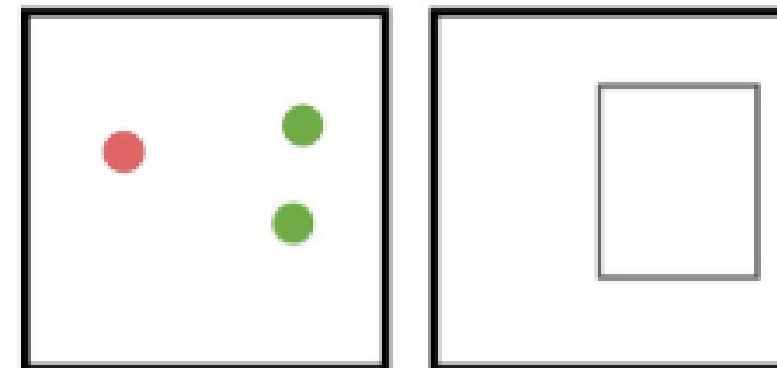
TP TRANSFORMER SEGMENTATION



FCN - Full classification
CLIPSeg - Prompt Segmentation

+

SAM - Point/Box Prompt

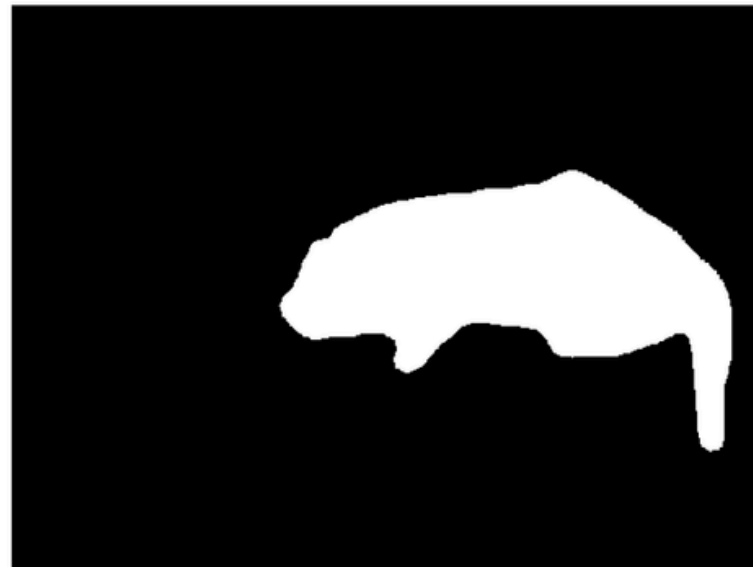


TP TRANSFORMER SEGMENTATION

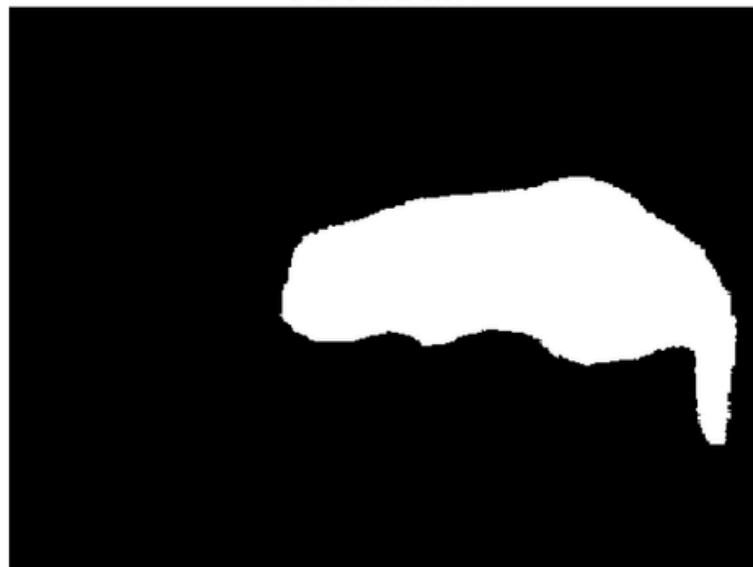
Original Image



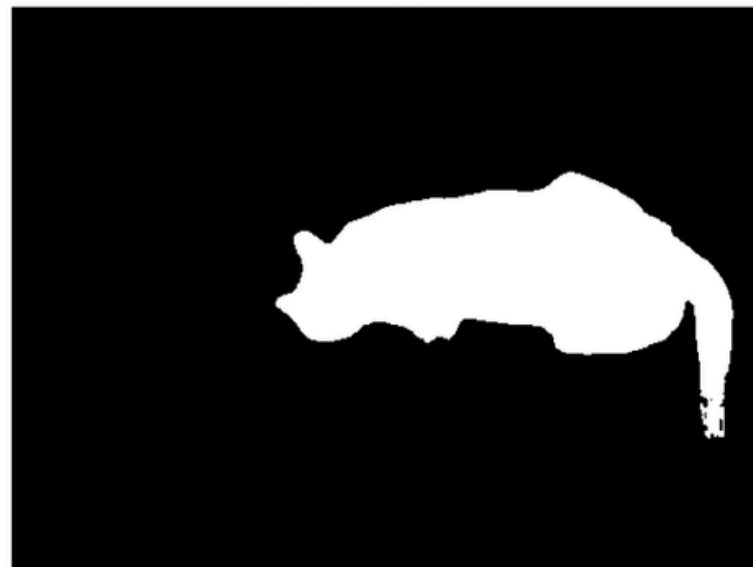
FCN Prediction
Dice Score: 0.962



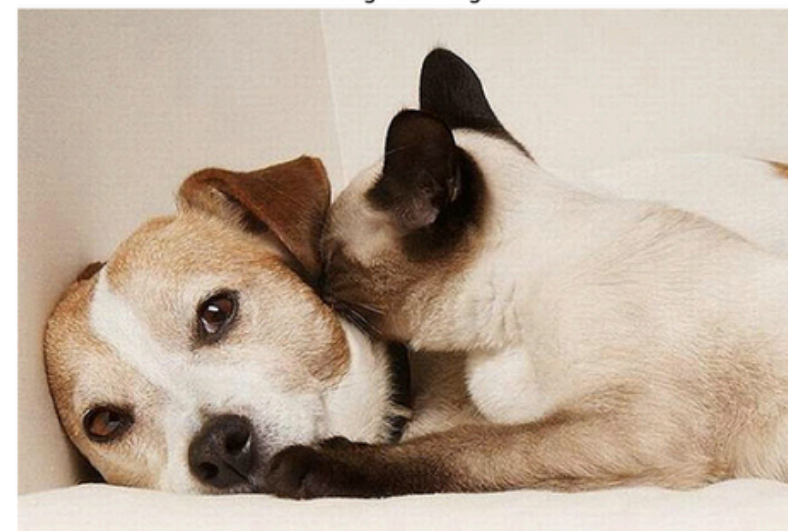
CLIPSeg Prediction
Dice Score: 0.956



SAM Prediction
Dice Score: 0.965



Original Image



FCN Prediction
Dice Score: 0.548



CLIPSeg Prediction
Dice Score: 0.735

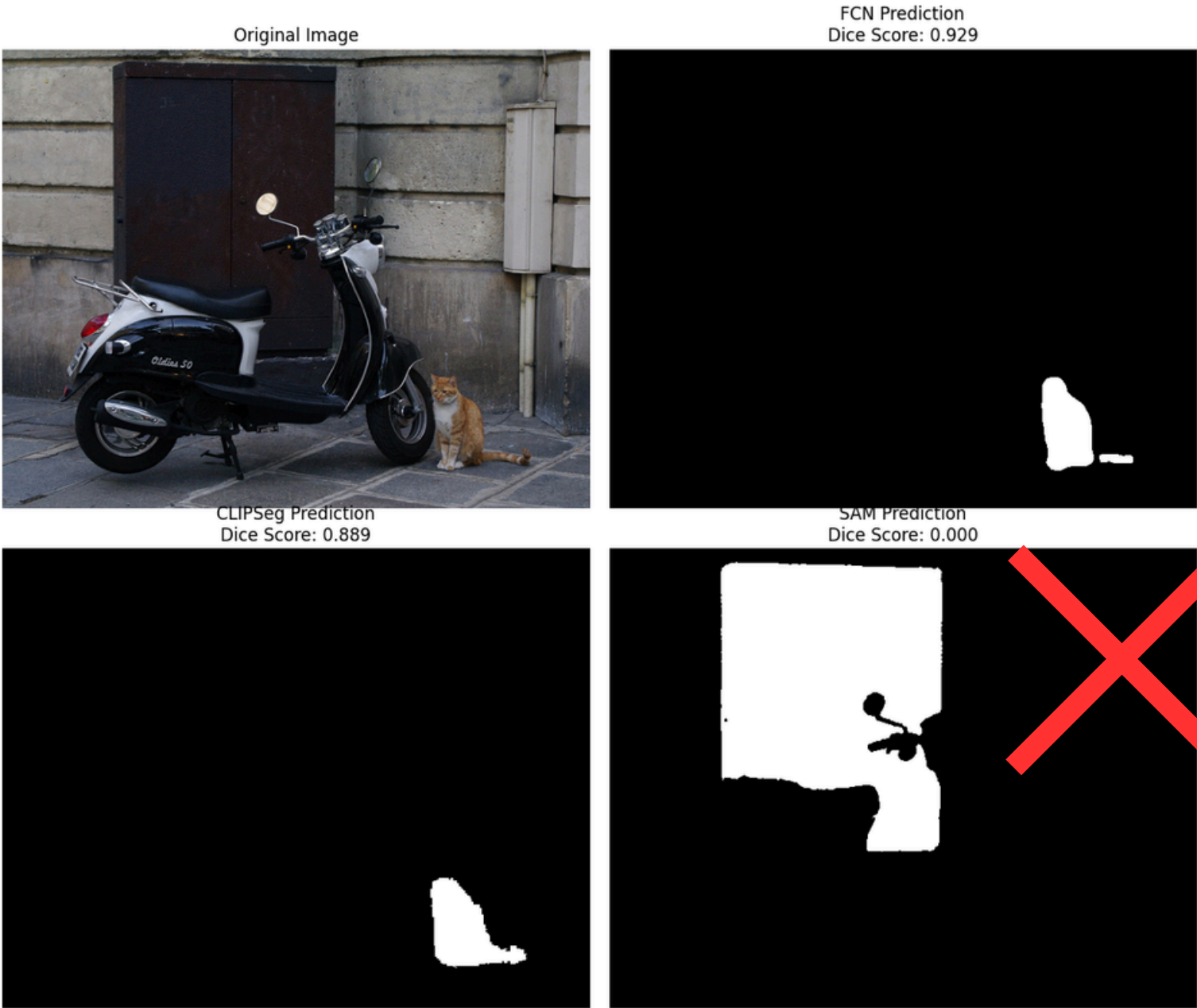


SAM Prediction
Dice Score: 0.752



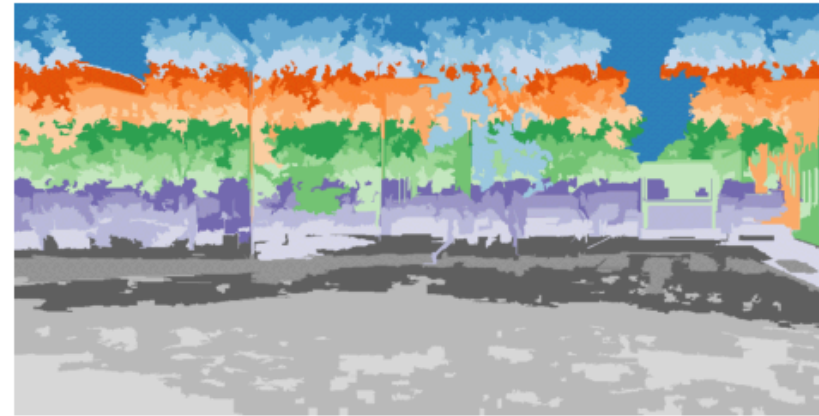
TP TRANSFORMER SEGMENTATION

Model	Mean Dice	Std
FCN	935,65	95,29
CLIPSeg	927,15	57,49
SAM	920,00	105,08

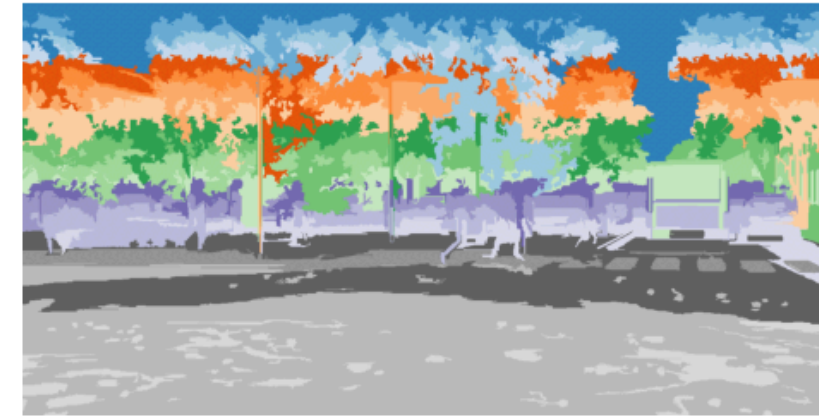




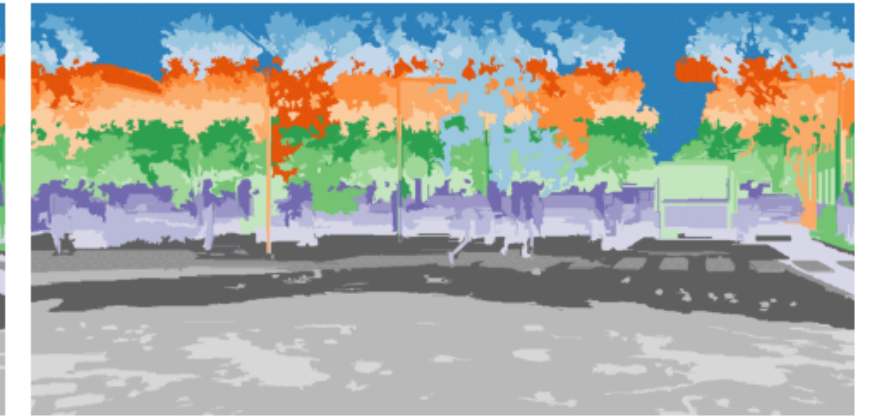
scale=100
sigma=0.5



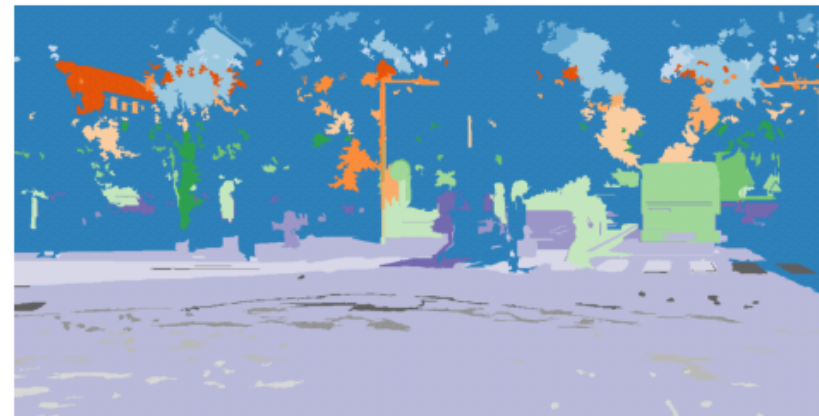
scale=100
sigma=0.9



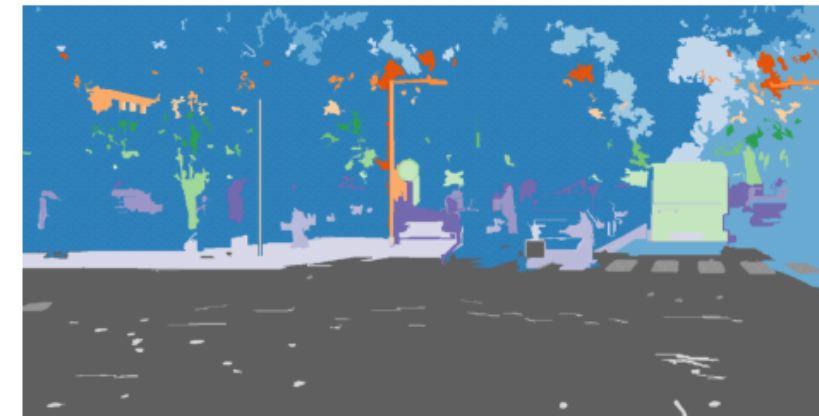
scale=100
sigma=0.99



scale=1000
sigma=0.5



scale=1000
sigma=0.9



scale=1000
sigma=0.99



scale=2000
sigma=0.5



scale=2000
sigma=0.9



scale=2000
sigma=0.99

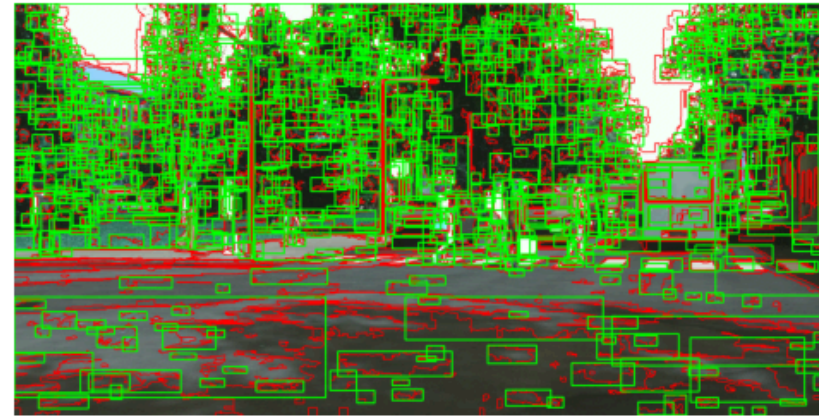


sigma – Standard deviation of a Gaussian filter applied before segmentation

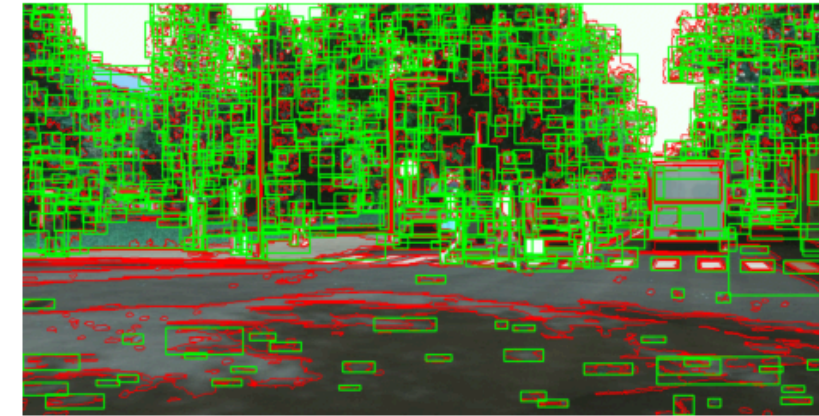
scale – Controls the threshold for merging regions



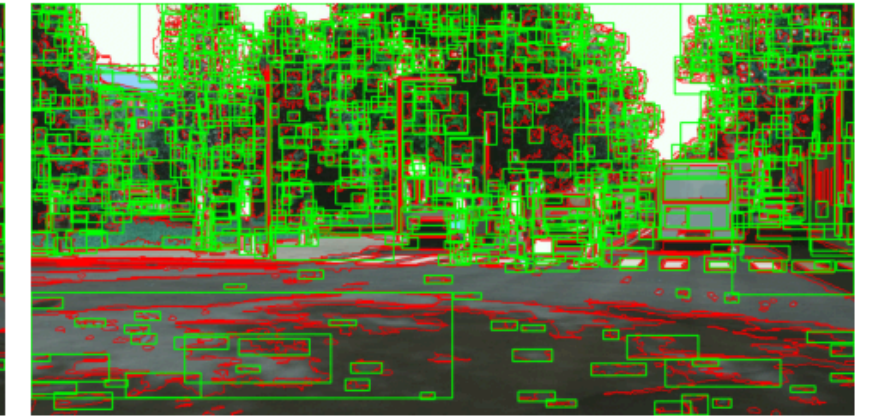
scale=100, sigma=0.5
Segments: 2507, Boxes: 974



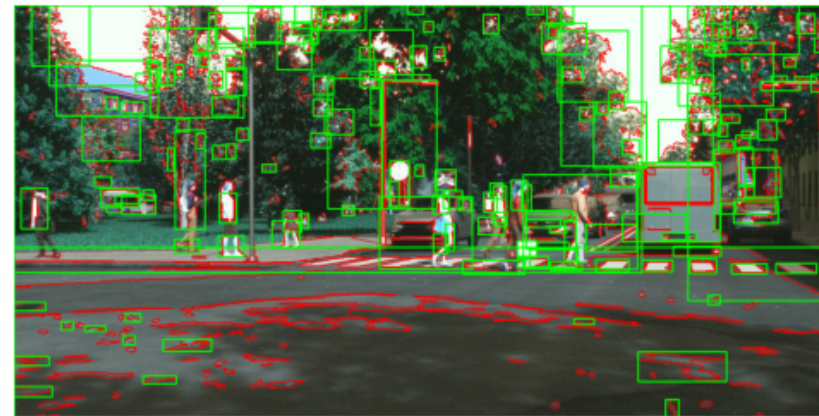
scale=100, sigma=0.9
Segments: 2253, Boxes: 821



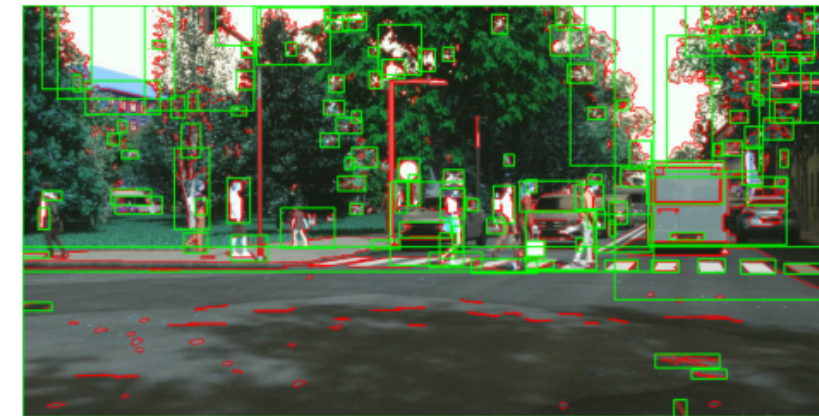
scale=100, sigma=0.99
Segments: 2158, Boxes: 790



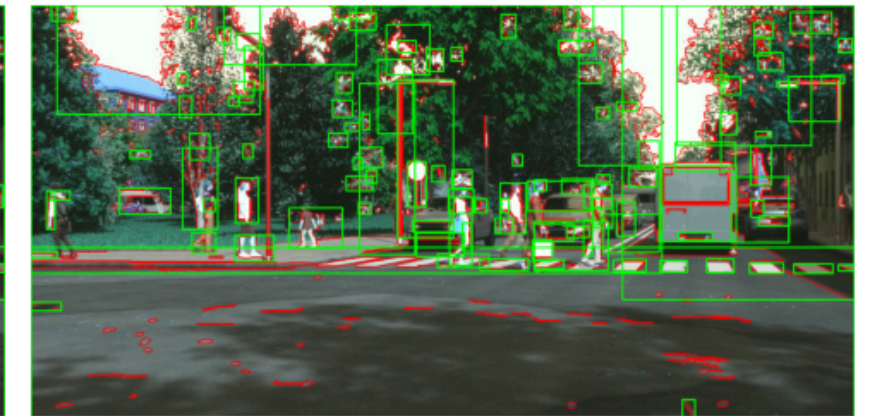
scale=1000, sigma=0.5
Segments: 504, Boxes: 174



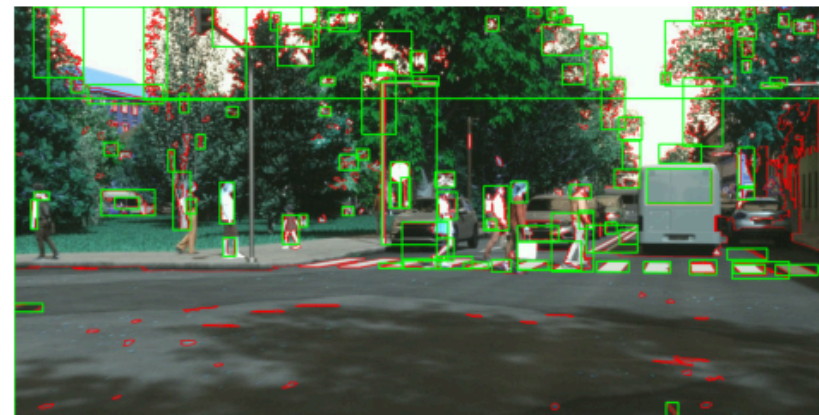
scale=1000, sigma=0.9
Segments: 329, Boxes: 131



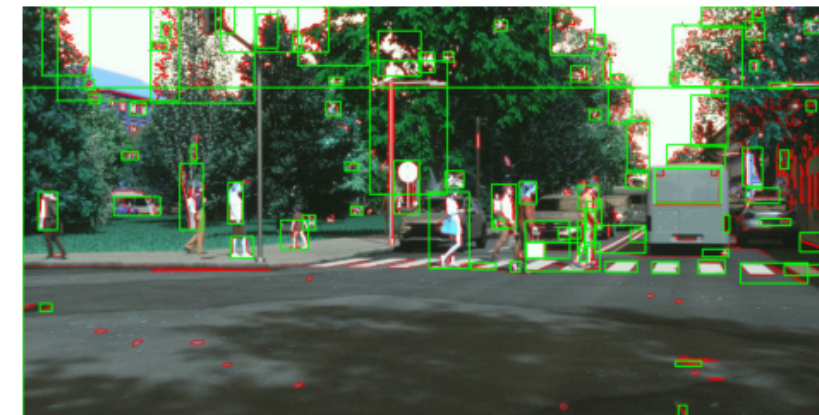
scale=1000, sigma=0.99
Segments: 325, Boxes: 114



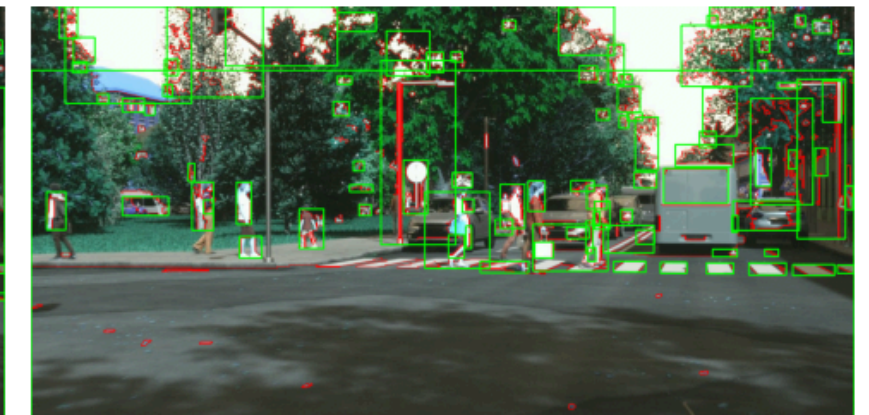
scale=2000, sigma=0.5
Segments: 264, Boxes: 101



scale=2000, sigma=0.9
Segments: 205, Boxes: 91



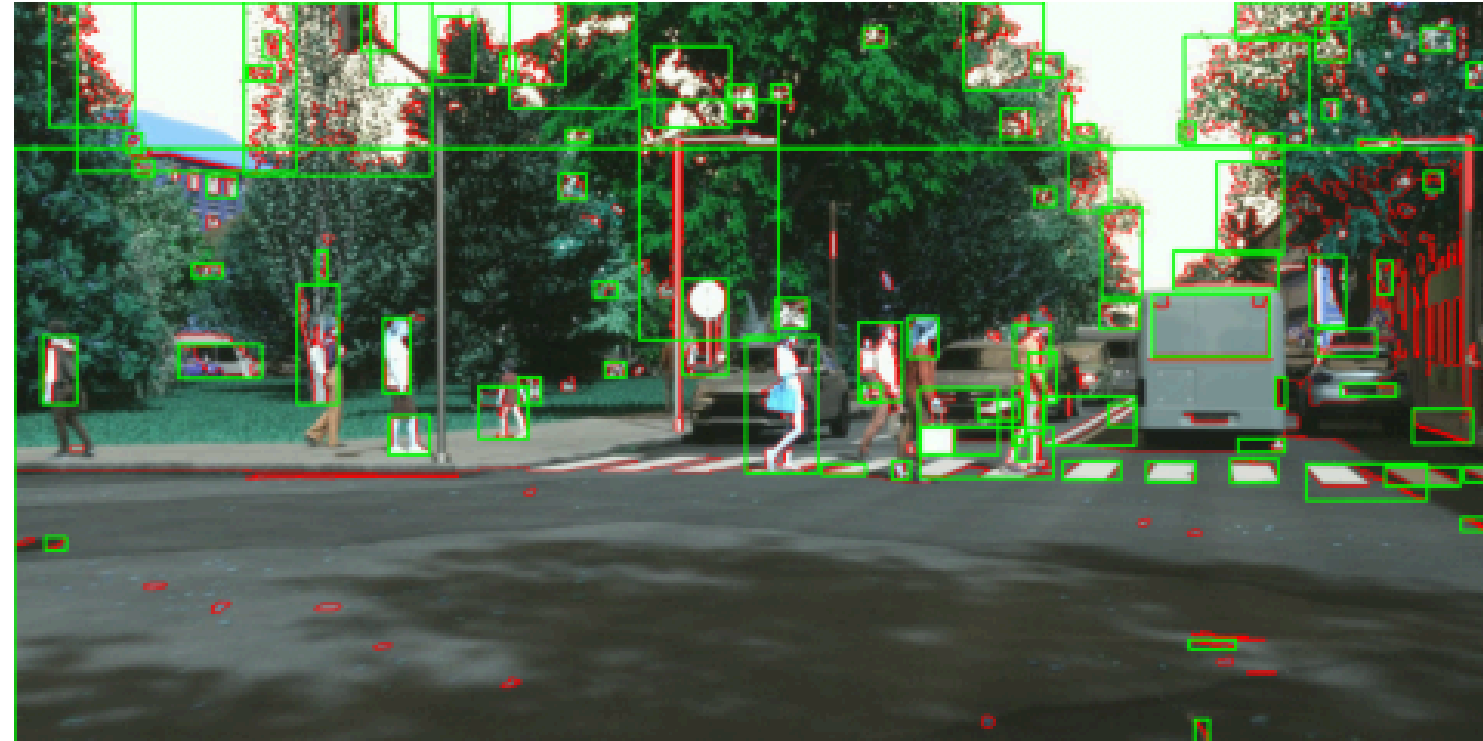
scale=2000, sigma=0.99
Segments: 200, Boxes: 94



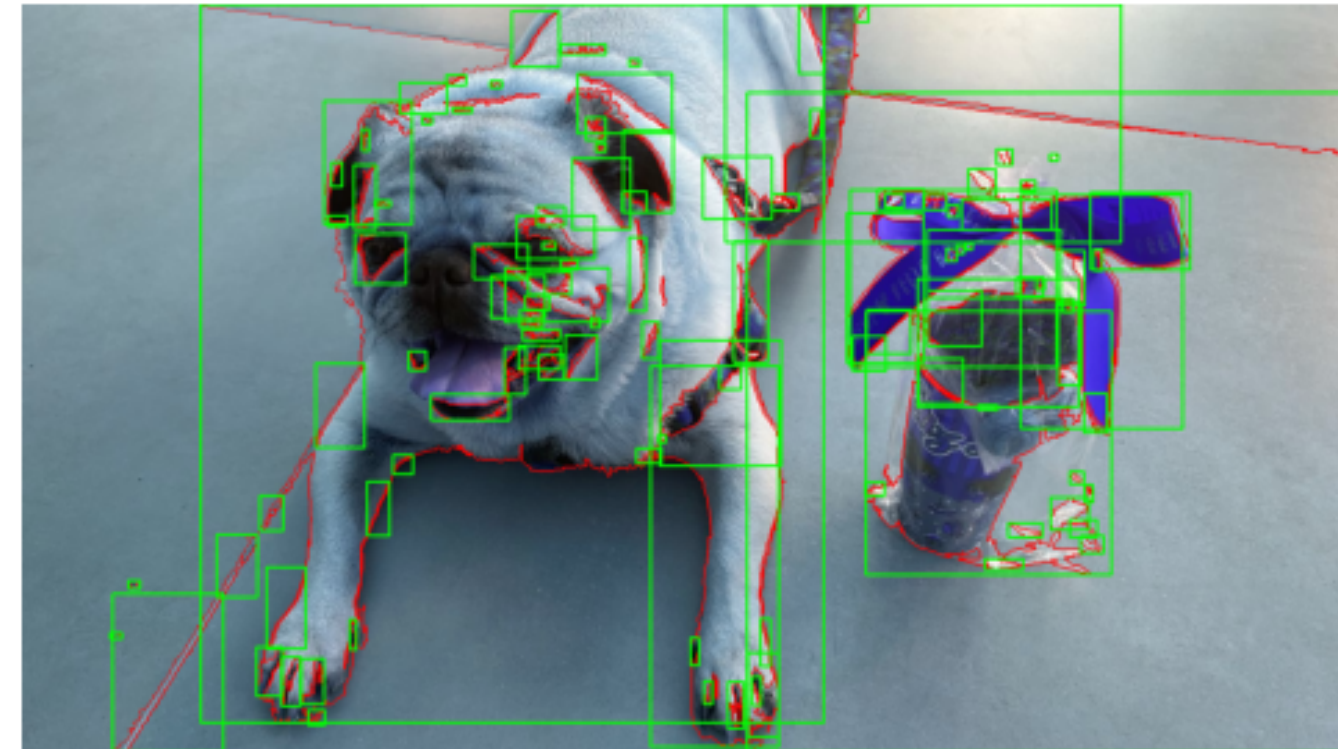
sigma – Standard deviation of a Gaussian filter applied before segmentation

scale – Controls the threshold for merging regions

scale=2000, sigma=0.9
Segments: 205, Boxes: 91

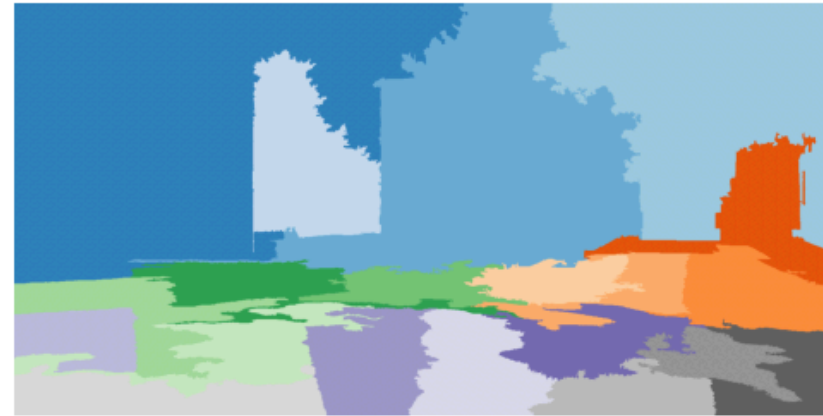


scale=1000, sigma=0.99
Segments: 138, Boxes: 114





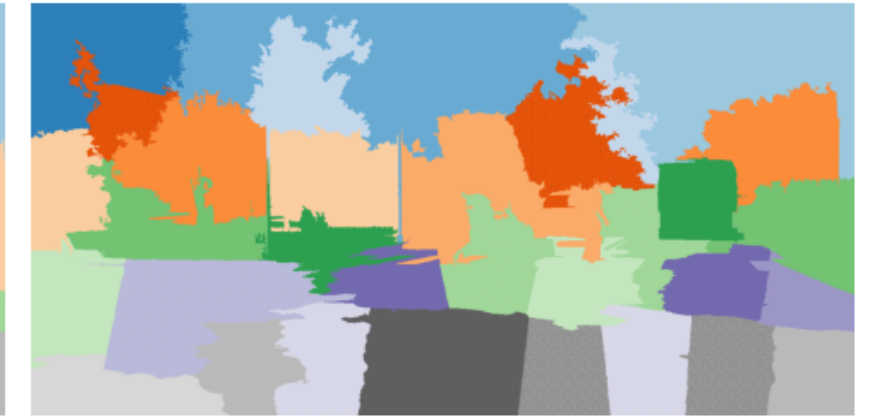
n_segments=50
compactness=5



n_segments=50
compactness=10



n_segments=50
compactness=20



n_segments=100
compactness=5



n_segments=100
compactness=10



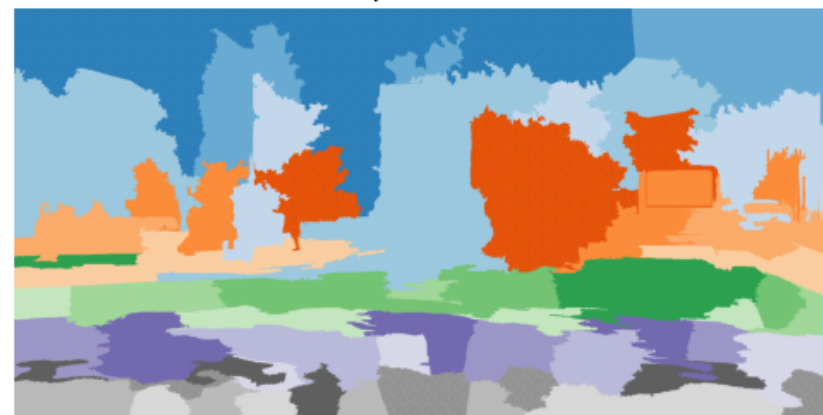
n_segments=100
compactness=20



n_segments – Desired number of superpixels

compactness – Spatial vs color trade off

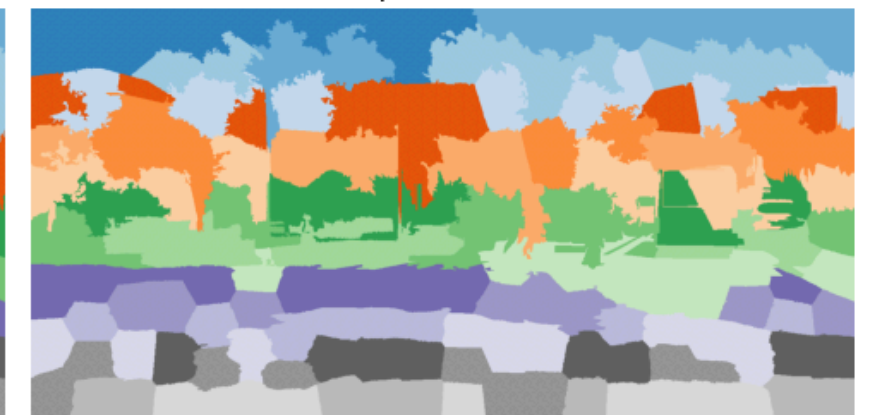
n_segments=200
compactness=5

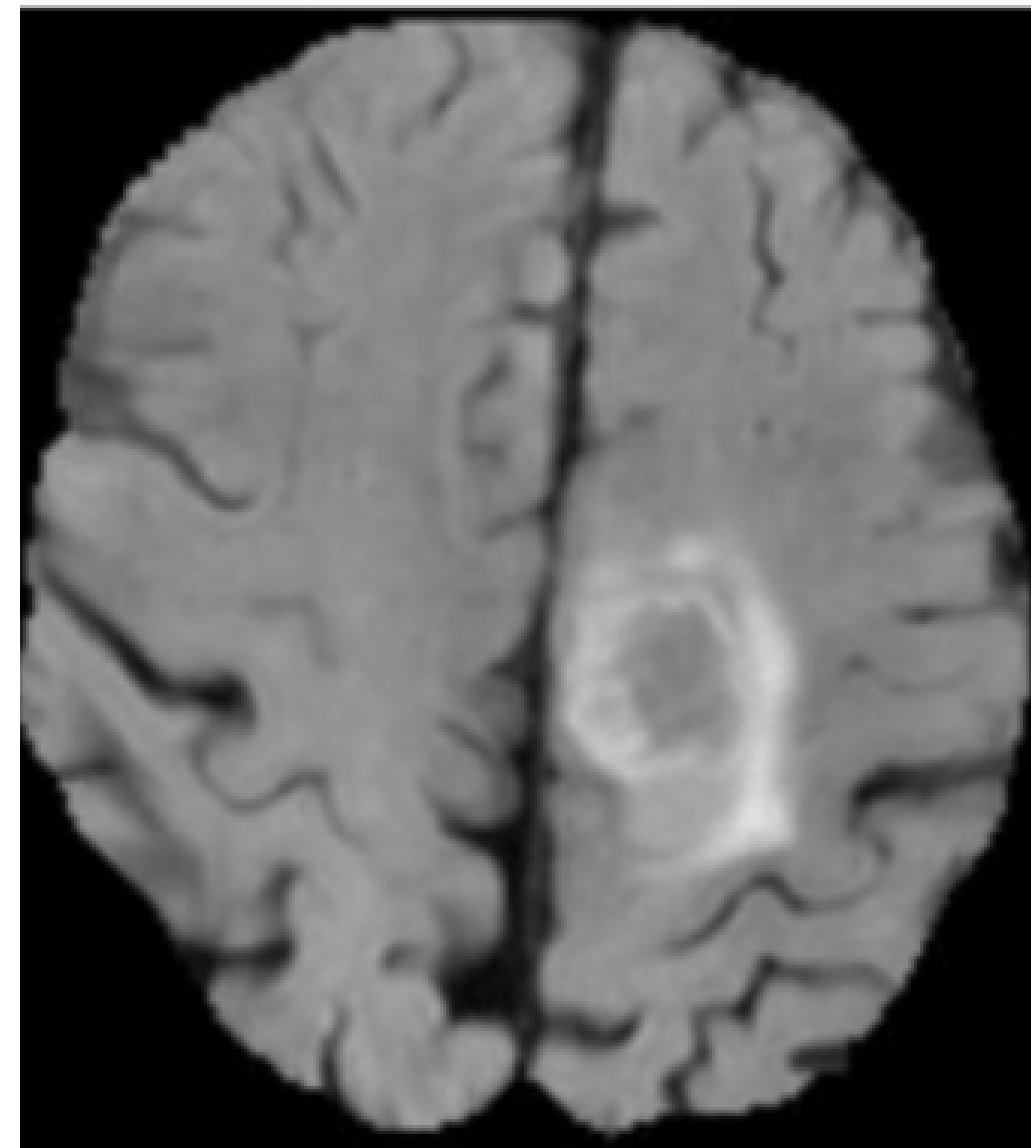


n_segments=200
compactness=10



n_segments=200
compactness=20



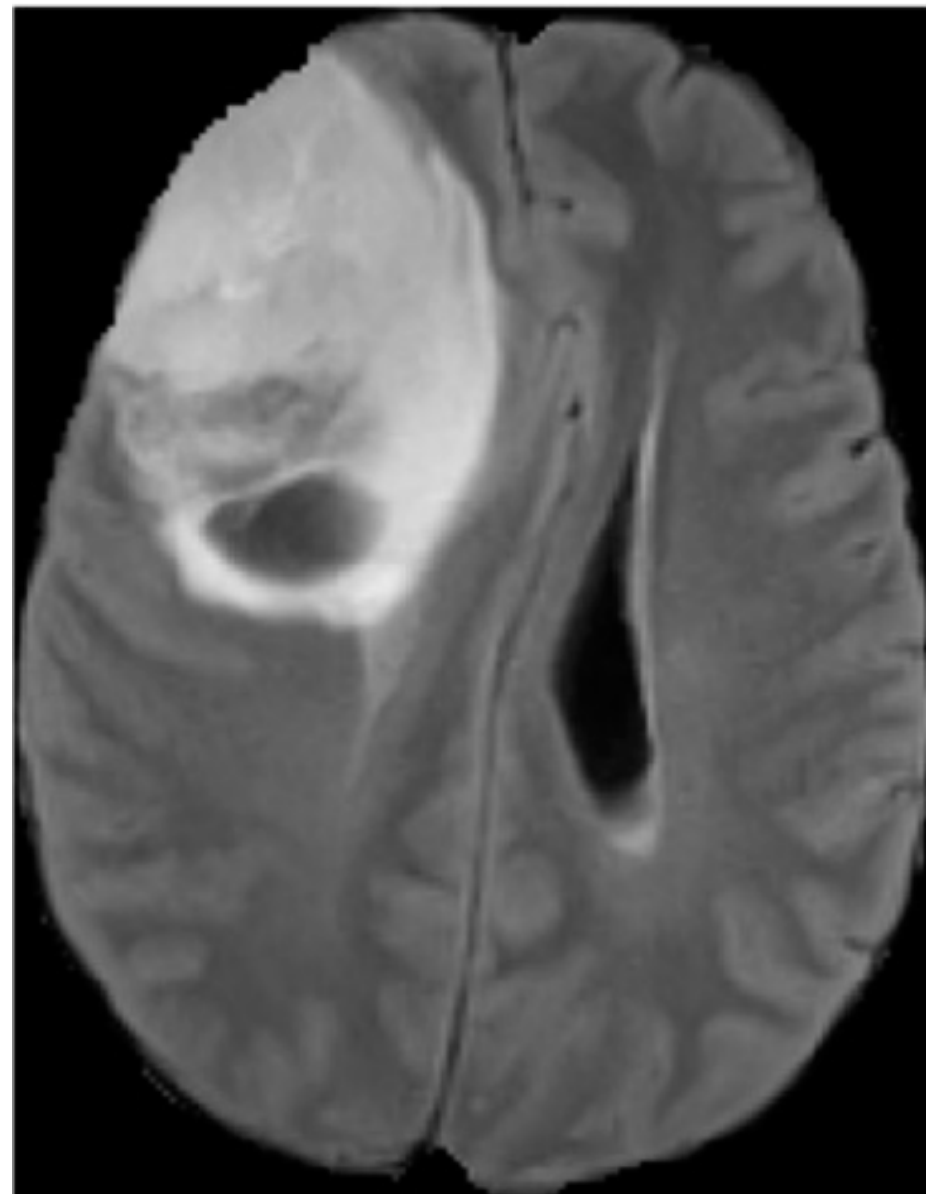


n_segments=50
compactness=10



n_segments=100
compactness=5





n_segments=100
compactness=5



n_segments=200
compactness=5

